How-To Library

Informatica

Implementing Big Data Management 10.2.2 HotFix 1 Service Pack 1 with Google Cloud Dataproc 1.3

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Abstract

You can integrate Informatica® Big Data Management 10.2.2 HotFix 1 Service Pack 1 with Google Dataproc 1.3 to run Big Data Management mappings and workflows in a Google cloud Hadoop implementation. This article describes how to integrate Big Data Management 10.2.2 HotFix 1 Service Pack 1 with the Dataproc cluster.

Supported Versions

- Big Data Management 10.2.2 HotFix 1 Service Pack 1

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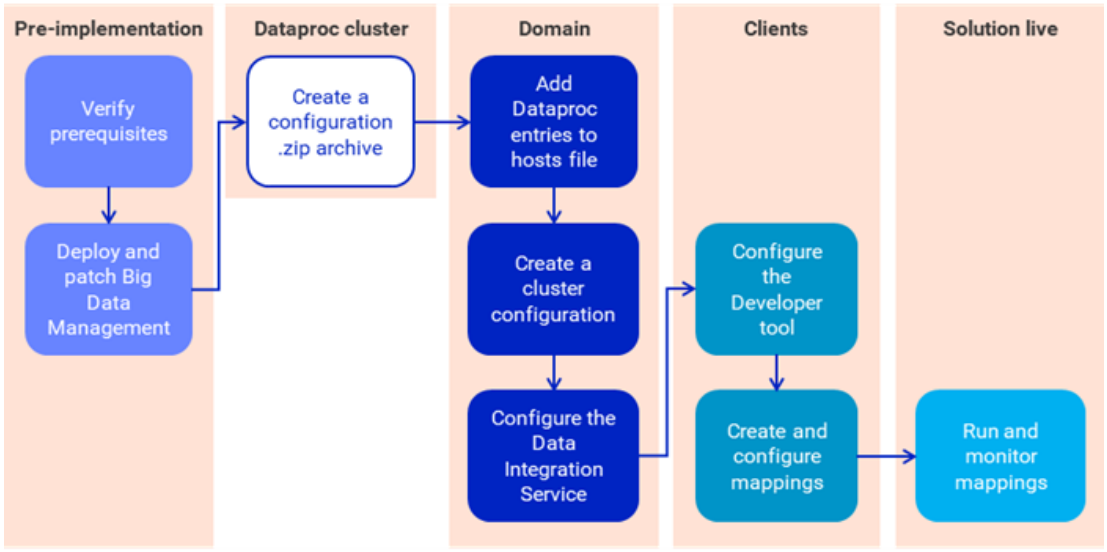
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Overview

Google Cloud Dataproc is a lightweight implementation of Hadoop and Apache Spark on the Google cloud platform. When you integrate Informatica big data products with Dataproc, you configure an on-premises Informatica domain to run jobs on the Dataproc cloud cluster.

When configuration is complete, you can run mappings from the Developer tool in the Dataproc environment.

The following image shows an overview of the integration process:



Pre-Implementation Tasks

Before you enable Big Data Management to run jobs on Google Cloud Dataproc, perform the following pre-implementation tasks:

- Verify prerequisites.
- Create the Big Data Management domain on-premises and apply the Dataproc implementation patch.

Verify Prerequisites

Verify the following prerequisites before you begin to configure Big Data Management for use with Dataproc:

- You have a Google Dataproc account with permissions to create resources, rules and policies.
- You have purchased a Big Data Management license.

Apply Informatica 10.2.2 HotFix 1 Service Pack 1

Before you apply Informatica 10.2.2 HotFix 1 Service pack 1, you must install Informatica version 10.2.2 HotFix 1 in an on-premise Linux environment and ensure that the Model Repository Service, Data Integration Service, and Metadata Access Service are running.

For more information, see the *Informatica Big Data Suite Installation and Configuration Guide*.

To apply the service pack, download the installer files from the Akamai link to your installation directory, update the `Input.properties` file, and run the installer.

You can run the installer to apply all product components or just a product component.

To roll back the service pack, set the rollback property in the `Input.properties` file, and then run the installer.

Service Pack Files

Informatica provides the service pack in .tar file and .zip file formats. After you download the service pack, extract the file contents. The service pack is available for Linux and Windows installations.

The service pack includes the following files:

Input.properties

Identifies the root directory of the Informatica installation to which you will install the service pack. You update the file with the directory path. The file also contains a rollback property that you can set if you decide to uninstall the service pack.

install.bat

Installs the service pack to the directory that you specify on a Windows machine. Find the file in the Windows installer.

install.sh

Installs the service pack to the directory that you specify on a Linux machine. Find the file in the Linux installer.

Download the Service Pack Files

Download one or more installer files to install a service pack.

To apply the service pack, you can download the server installer, Informatica Developer installer, and command line utilities package.

Download one of the following packages:

- informatica_10221SP1_server_linux-x64.tar
  Contains updates for Redhat Enterprise Linux installations.
- informatica_10221SP1_server_suse11-x64.tar
  Contains updates for SUSE Linux Enterprise Server installations.

Download the following package for Informatica Developer:

- informatica_10221SP1_client_winem-64t.zip
  Contains updates for Developer tool installation.

Download one of the following packages for command line utilities:

- informatica_10221SP1_cmd_utilities_linux-x64.zip
  Contains updates for Redhat Enterprise Linux installations.
- informatica_10221SP1_cmd_utilities_suse11-x64.zip
  Contains updates for SUSE Linux Enterprise Server installations.

Update the Input.properties File

The *Input.properties* file includes properties that identify the Informatica installation and define the action taken when you run the service pack installer file. Update the properties before you install the service pack. Update the file in each service pack that you download.

1. Extract the service pack file archive.
2. Find the *Input.properties* file in the extracted service pack file.
3. Update the *DEST_DIR* property in the file with the path to the Informatica root directory.
   - On a Linux machine, set the path in the following format:
     ```
     DEST_DIR=/home/infauser/<version number>
     ```
   - On a Windows machine, set the path in the following format:
     ```
     DEST_DIR=C:\\Informatica\\<version number>
     ```
4. Configure the value of the ROLLBACK property. You can apply or roll back the service pack for all product components or to specific component. To install the service pack, retain the default value of **0**. To roll back the service pack, set the value to **1**.

When you install the service pack, the installer applies all the components, by default. To apply a specific component, remove the comment tag (#) associated with the component that you want to apply.

For Big Data Management component, set BDM_ONLY to 1. For Enterprise Data Catalog, set EDC_ONLY to 1. For Enterprise Data Preparation, set EDP_ONLY to 1.

You must apply individual components for a customized product application in the following order:

1. Big Data Management
2. Enterprise Data Catalog
3. Enterprise Data Preparation

You must roll back the components for a customized product application in the following order:

1. Enterprise Data Preparation
2. Enterprise Data Catalog
3. Big Data Management

5. Save and close the file.

Run the Installer

Run the installer file to install the service pack.

1. Close all Informatica applications and stop all Informatica services.
2. Find the installer file in the service pack files and extract the file.
   - For Linux systems, the installer file is *install.sh*.
   - For Windows systems, the installer file is *install.bat*.

Create a Configuration Files .zip Archive

A set of configuration files on the Dataproc cluster contains settings that configure Hadoop functionality. To enable the Informatica domain to access the Dataproc cluster, you must create a .zip archive with these configuration files and copy it to the Informatica domain node.

1. Log in to the cluster master node and browse to the `hadoop/conf` directory.
2. Add the following property to the yarn-site.xml file:

   ```
   yarn.resourcemanager.address=<namenodehost>:8032
   ```

3. Add the following property to the hdfs-site.xml file:

   ```
   <property>
     <name>dfs.permissions.enabled</name>
     <value>true</value>
   </property>
   ```

4. Copy the following files:
   - hdfs-site.xml
   - yarn-site.xml
   - mapred-site.xml
   - core-site.xml
5. Browse to the `/hive/conf` directory and copy the hive-site.xml file.
6. Create a .zip archive that contains all five configuration files.
7. Copy the .zip archive file to a directory on the Informatica domain machine.

You use the .zip archive file later to create a cluster configuration object on the domain.

Configuring the Informatica Domain

Configure the Informatica domain to connect to the Google Dataproc cluster and run mappings and workflows.

Perform the tasks in this section to configure the domain:

- Step 1: Add Dataproc entries to the domain etc/hosts file.
- Step 2: Create a cluster configuration for the domain.
- Step 3: Configure execution options on the Data Integration Service.

Add Dataproc Entries to the Domain etc/hosts File

To enable Informatica to access the Dataproc cluster, edit the `/etc/hosts` file on the domain host and on client machines.

Enter the IP address, DNS name, and DNS short name for each data node on the cluster.

For example:

```
10.20.30.40 dataprocABC-m.c.MyUsernameINFA54321.internal dataprocABC-m
10.20.30.44 dataprocABC-w-0.c.MyUsernameINFA54321.internal dataprocABC-w-0
10.20.30.43 dataprocABC-w-1.c.MyUsernameINFA54321.internal dataprocABC-w-1
10.20.30.42 dataprocABC-w-2.c.MyUsernameINFA54321.internal dataprocABC-w-2
```

Important: Edit the hosts file with cluster node information on the following nodes:

- Each machine that hosts the Data Integration Service
- Each machine that hosts the Developer tool

Create a Cluster Configuration

After the Hadoop administrator prepares the cluster for import, the Informatica administrator must create a cluster configuration.

A cluster configuration is an object in the domain that contains configuration information about the Hadoop cluster. The cluster configuration enables the Data Integration Service to push mapping logic to the Hadoop environment. Import configuration properties from the Hadoop cluster to create a cluster configuration.

The import process imports values from *-site.xml files into configuration sets based on the individual *-site.xml files. When you perform the import, the cluster configuration wizard can create Hadoop, HDFS, and Hive connection to access the Hadoop environment. If you choose to create the connections, the wizard also associates the cluster configuration with the connections.

For more information about the cluster configuration, see the *Big Data Management Administrator Guide*.

Importing a Cluster Configuration from a File

You can import properties from an archive file to create a cluster configuration.

A cluster configuration is an object in the domain that contains configuration information about the Hadoop cluster. The cluster configuration enables the Data Integration Service to push mapping logic to the Hadoop environment.

The cluster configuration contains properties for the cluster type and version. You can edit the version property to any supported version. After you change the version, you must restart the Data Integration Service.

1. From the **Connections** tab, click the **ClusterConfigurations** node in the Domain Navigator.

2. From the Actions menu, select **New** > **Cluster Configuration**.

   The **Cluster Configuration** wizard opens.

3. Configure the following properties:

Property	Description
Cluster configuration name	Name of the cluster configuration.
Description	Optional description of the cluster configuration.
Distribution type	The cluster Hadoop distribution type. Select Google Dataproc.
Distribution version	Version of the Hadoop distribution. Each distribution type has a default version. This is the latest version of the Hadoop distribution that Big Data Management supports. You can edit the property to choose any supported version. Restart the Data Integration Service for the changes to take effect.
Method to import the cluster configuration	Choose **Import from file** to import properties from an archive file.
Upload configuration archive file	Click **Browse** to locate and select the .zip archive that contains the *-site.xml cluster configuration files.
Create connections	Choose to create Hadoop, HDFS, Hive, and HBase connections. If you choose to create connections, the Cluster Configuration wizard associates the cluster configuration with each connection that it creates. If you do not choose to create connections, you must manually create them and associate the cluster configuration with them. **Important:** When the wizard creates the Hive connection, it populates the Metadata Connection String and the Data Access Connection String properties with the value from the hive.metastore.uris property. If the Hive metastore and HiveServer2 are running on different nodes, you must update the Metadata Connection String to point to the HiveServer2 host.

4. Click **Next** and verify the cluster configuration information on the summary page.

Configure the Data Integration Service

Use the Administrator tool to configure domain service properties.

Perform the following steps to configure the data information service to use the Dataproc cluster:

1. In the domain navigator, select the Data Integration Service.
2. Click the **Edit** icon to edit Execution Option properties for the Data Integration Service.
   The following image shows that the **Edit** icon resembles a pencil:



3. Configure the Hadoop Staging User property as shown in the following table:

Property	Description
Hadoop Staging User	User account to use to communicate with the Dataproc cluster. Use the same user account ID that you use to login to Dataproc cluster nodes.

4. Save the changes to Data Integration Service properties and recycle the Data Integration Service.

Enable Access to Google Cloud Sources

You can use Google Cloud resources as mapping sources. You can enable access to Google Cloud storage by saving account credentials in a JSON file on each Data Integration Service machine.

1. Create service account credentials in JSON file format.

   Set permissions for the account to access the Google Cloud resources that act as data sources for mappings. To create the JSON credentials file, see Google Cloud documentation.

2. Save the JSON file on the machine that hosts the Data Integration Service. Set full permissions to enable the Data Integration Service to access the file.
3. In the Administrator tool, select the Data Integration Service and edit the Environment Variables properties to point to the JSON file.

   Add the following key-value pair to the environment variables:

   Key: GOOGLE_APPLICATION_CREDENTIALS

   Value: `<path to the JSON file>`

4. Recycle the Data Integration Service.

Note: Perform these steps on each machine that hosts the Data Integration Service.

Configure the Developer Tool to Use the Dataproc Cluster

Perform this step on each machine that hosts the Developer tool.

1. Browse to the following path on the Developer tool machine:

   `<Informatica home directory>\<version>\clients\DeveloperClient`

2. Open the `developerCore.ini` file for editing.
3. Add the following line to the file:

   `-DINFA_HADOOP_DIST_DIR=Hadoop\Dataproc_1.3`

4. Save and close the file, and then restart the Developer tool.

Configure Mappings to Run on Dataproc

To run mappings on the Dataproc cluster, configure mappings with the following properties:

1. In the **Parameters** section, create a parameter with the values shown in the following table:

Name	Type	Precision	Scale	Default Value
pushdownConnection	Connection	(blank)	(blank)	Use the dropdown control to select the Hadoop connection that the cluster configuration created.

2. In the **Run-Time** section, choose the following values:
   - Under Validation Environments, select Spark.
   - For Execution Environment, select Hadoop.
   - Under the Hadoop section, for the Connection property, click the right side of the Value cell to see the dropdown menu, and choose the pushdownConnection parameter.

Monitoring Big Data Management Jobs

To monitor the status of Big Data Management jobs on the Dataproc cluster, access the Hadoop Resource Manager.

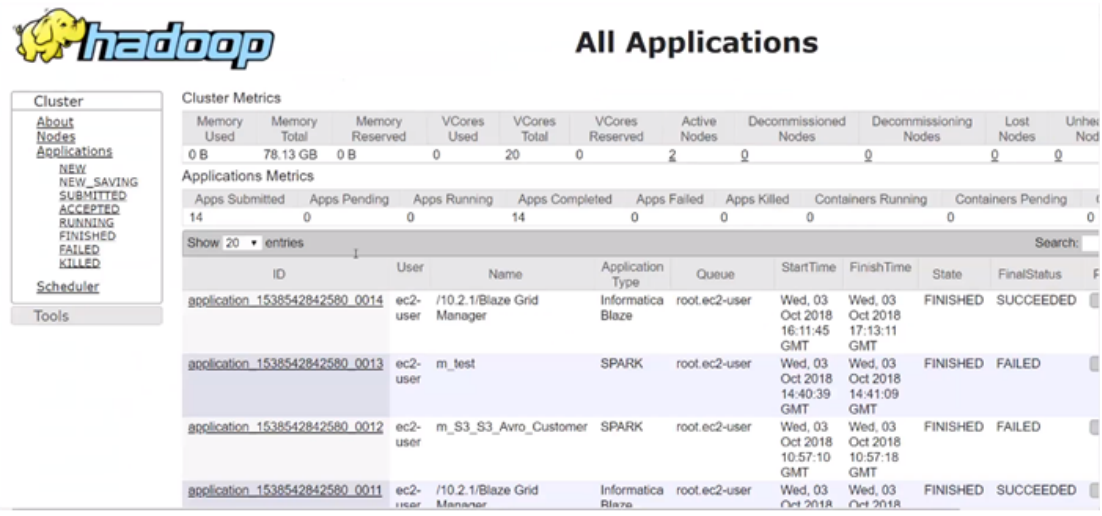
To access the Hadoop Resource Manager, browse to the following URL:

```
http://<Master_node_name>:8088
```

where the Master_node_name is the node name or IP address of the Dataproc master node. For example:

```
http://dataproc123-m:8088
```

The following image shows the Hadoop resource manager:



Troubleshooting Mapping Failures on the Dataproc Cluster

This section provides information on troubleshooting common error messages and limitations that you might encounter when you run mappings on a Dataproc cluster.

When you run mappings on a Dataproc cluster to read or write data to SerDe-backed Hive tables, the mapping fails with the java.lang.ClassNotFoundException: org.apache.hive.hcatalog.data.JsonSerDe error.

Add the following property in the `hive-site.xml` on all the nodes of the Dataproc cluster and restart the Hive server:

```
<property>
  <name>hive.aux.jars.path</name>
  <value>file:///usr/lib/hive/lib/hive-hcatalog-core-<version>.jar</value>
</property>
```

When you run mappings on a non-VPN Dataproc cluster, the mappings fail.

Configure the following properties in the `hdfs-site.xml` on all the nodes of the Dataproc cluster:

```
<property>
  <name>dfs.namenode.rpc-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.namenode.servicerpc-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.namenode.http-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.namenode.https-bind-host</name>
  <value>0.0.0.0</value>
</property>
<property>
  <name>dfs.client.use.datanode.hostname</name>
  <value>true</value>
</property>
```

Known Issues

- When you run a mapping on a Dataproc cluster to read data from or write data to a .csv file in Microsoft Azure Blob Storage, the mapping fails with the following error:
  `SEVERE: [Pre_Spark_Task_Command_1] [HadoopFSRmRfTask]java.lang.RuntimeException: java.lang.ClassNotFoundException: Class org.apache.hadoop.fs.azure.NativeAzureFileSystem not found`
- When you run a mapping on a Dataproc cluster to read or write complex files and you enable compression, the mapping fails with the following error:
  `Error: [Exception during complex file read. Root cause: java.lang.UnsatisfiedLinkError: org.xerial.snappy.SnappyNative.uncompressedLength(Ljava/nio/ByteBuffer;II)I ].].`
- When you run a Sqoop mapping on a Dataproc cluster to read data from or write data to Microsoft SQL Server, the mapping fails with the following error:
  `java.lang.RuntimeException: com.microsoft.sqlserver.jdbc.SQLServerException: The driver could not establish a secure connection to SQL Server by using Secure Sockets Layer (SSL) encryption. Error: "Socket is closed"`
- You cannot run mappings on a Dataproc cluster to read data from or write data to an HBase table. Google Dataproc does not support HBase service.

Author

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